THE CHAI STRAINER AND THE SALT PAN

KABIR AND ANYA'S RAINY DAY CHALLENGE

After heavy rain, Kabir and Anya's cricket ground was a muddy mess! Their water bucket was mixed with dirt and sand. Can you help them get clean water, clean sand, and salt back?

WHAT IS A MIXTURE?

When two or more things are mixed but can still be separated!

Examples: sugar in water, sand in water, tea with leaves

Soluble = can dissolve • Insoluble = cannot dissolve

HOW FILTRATION WORKS
THE BARRIER RULE

RESIDUE

The solid bits that are too big to pass through the filter get trapped and stay behind. This is called the *RESIDUE*.
Example: Tea leaves caught in the chai strainer!

FILTRATE

The liquid that passes through the tiny holes in the filter is called the *FILTRATE*.
Example: The smooth liquid tea that flows into your cup!

INDIAN EXAMPLE: THE CHAI STRAINER

"Chai-ni" (tea strainer) lets liquid tea pass through but catches the tea leaves behind!

🍵 Residue = Tea leaves (trapped in strainer) |
Filtrate = Liquid tea (passes through)

HOW EVAPORATION WORKS!

When water is heated, it turns into steam and floats away — leaving solids behind! Dissolved solids like salt are too tiny for a filter, so we use heat instead. The water disappears, and only the solid remains.

⚠️ Safety Note: Always do heating experiments with adult supervision!

INDIAN EXAMPLE: THE SALT PAN

In Gujarat and Tamil Nadu, sunlight evaporates seawater in vast salt pans. As the water disappears into steam, salt crystals are left behind — this is natural evaporation! Salt farmers then collect the white salt crystals from the pans.

WHEN WE NEED BOTH! 🔬

STEP 1: FILTRATION

Pour the sand + salt + water mixture through a filter cloth.
✅ Sand gets TRAPPED — this is the Residue!
💧 Salt water passes through — this is the Filtrate!

STEP 2: EVAPORATION

Heat the salt water (filtrate) gently in a pan.
☁️ Water turns to steam and disappears!
✅ Salt CRYSTALS are left behind — just like the Salt Pans of Gujarat!

GUIDED PRACTICE: FILTERING RIVER WATER

Kabir folds a piece of Suti Kapda (white cotton cloth) and places it over an empty bucket. He slowly pours muddy river water through the cloth and watches carefully...

What gets TRAPPED in the cloth?

- Silt (fine mud particles)
- Dirt and small pebbles
- Tiny bits of leaves and debris

What PASSES THROUGH the cloth?

- Clear, cleaner water (the Filtrate!)

The cotton cloth acts like a filter — its tiny holes let water through but block bigger solid particles. The trapped dirt is the Residue. The clean water collected below is the Filtrate!

SALT HARVESTING AT RANN OF KUTCH 🧂

Salt farmers at the Rann of Kutch collect salt after the bright sun evaporates seawater. Step 1: Seawater flows into wide, shallow salt pans. Step 2: The hot sun heats the water day after day. Step 3: Water turns into steam and rises into the air!

Step 4: White salt crystals are left behind in the pans. Step 5: Farmers rake and collect the salt. This is natural evaporation at work! The dissolved salt cannot escape as steam — only water can. Gujarat's Rann of Kutch is one of the world's largest salt deserts!

MATCH THE MIXTURE TO THE METHOD!

FILTER 🫙
PEBBLES + WATER
TURMERIC + WATER

EVAPORATE 🔥
SUGAR + WATER
SALT + WATER

SUMMARY: REMEMBER THESE RULES!

FILTER IT! 🧹

Insoluble = Filter it!
Big particles get trapped.
Some mixtures need BOTH steps — filter first, then evaporate!
Key Words: Mixture • Filtration • Residue • Filtrate

EVAPORATE IT! ☀️

Soluble = Evaporate it!
Liquid disappears, solid stays.
Salt is soluble — you cannot catch it with a filter!
Key Words: Soluble • Insoluble • Evaporation • Crystals

QUICK QUIZ TIME!

QUESTION 1

What does a filter catch? What is the solid left behind called?

RESIDUE! ✓

QUESTION 2

What happens to water during evaporation? Can salt be removed by filtering?

STEAM! NO! ✓

YOUR HOME MISSION!

WATCH YOUR PARENTS MAKE CHAI (TEA) OR FILTER WATER AT HOME. BE A SCIENCE DETECTIVE! 🔍

✅ SPOT THE RESIDUE – WHAT GETS TRAPPED BEHIND? (TEA LEAVES, DIRT, SILT)
✅ SPOT THE FILTRATE – WHAT PASSES THROUGH? (LIQUID TEA, CLEAN WATER)
✅ CAN YOU FIND EVAPORATION HAPPENING TOO? (STEAM FROM THE POT!)
🌟 SHARE WHAT YOU DISCOVER IN OUR NEXT CLASS – WE CAN'T WAIT TO HEAR YOUR FINDINGS!

THANK YOU!

KEEP EXPLORING SCIENCE!